>> Intelligent Systems – KGs and Recap <<

Rule-based Reasoning and Knowledge Graphs

Selection of exercises extra preparation for exam

NOTE: this set of exercises is selected based on our assumptions about topics that need more attention after teaching WC1-WC3, and does not reflect the structure of the exam. There is no time to discuss all topics, but in case you have questions about topics other than in this set of exercises, do not hesitate to ask them.

Try to make all exercise before you come to WC. By this means you will be aware of your own (lack of) knowledge about the topics. Good luck!

Exercises Knowledge Graphs

Exercise KG1: Rule-based reasoning: Recall exercise 13 from the previous working group. Prove or disprove whether a knowledge base KB={**C -> -(A v -B), A}** entail **-C** or not, but now using Forward Checking (FC) rule based reasoning. Give the rules that you apply, and write down the closure at the end:

$$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha) \quad \text{commutativity of } \wedge$$
$$(\alpha \vee \beta) \equiv (\beta \vee \alpha) \quad \text{commutativity of } \vee$$
$$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma)) \quad \text{associativity of } \wedge$$
$$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma)) \quad \text{associativity of } \vee$$
$$\neg(\neg\alpha) \equiv \alpha \quad \text{double-negation elimination}$$
$$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha) \quad \text{contraposition}$$
$$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta) \quad \text{implication elimination}$$
$$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) \quad \text{biconditional elimination}$$
$$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta) \quad \text{de Morgan}$$
$$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta) \quad \text{de Morgan}$$
$$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) \quad \text{distributivity of } \wedge \text{ over } \vee$$
$$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) \quad \text{distributivity of } \vee \text{ over } \wedge$$

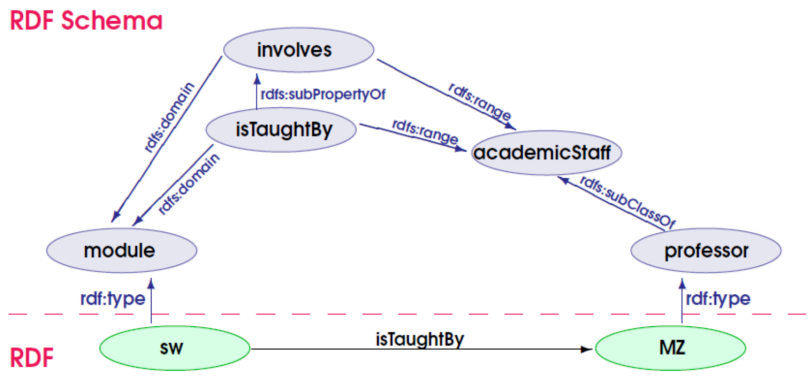
$$\frac{\alpha \Rightarrow \beta, \alpha}{\beta}$$

$$\frac{\alpha \wedge \beta}{\alpha}$$

$$\frac{\alpha \Leftrightarrow \beta}{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)}$$

Exercise KG2: Rule-based reasoning: Look at the following variant of the previous exercise. Prove or disprove whether a knowledge base KB={**C -> -(A v -B), C}** entail **-A** or not, using Forward Checking (FC) rule based reasoning. Give the rules that you apply, and write down the closure at the end:

Exercise KG3: Knowledge Graphs: Write down the following Knowledge Graph as a set of triples. Discuss whether there are any non-trivial triples that are entailed.



Exercise 3: Reasoning with RDFS rules: Suppose the following rule (only) is giving.

rdfs9: xxx rdfs:subClassOf yyy AND zzz rdf:type xxx ⇒ zzz rdf:type yyy .

and an RDFS graph:

{ (ex:A rdfs:subClassOf ex:D), (ex:D rdfs:subClassOf ex:C), (ex:B rdfs:subClassOf ex:D), (ex:i1 rdf:type ex:A), (ex:i2 rdf:type ex:B)}

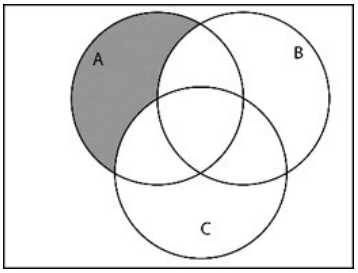
Which triples can be derived?

Exercises Recap

Exercise 1: Sets: Which is an example of disjoint sets?

A. {multiples of two} and {multiples of three}
B. {whole numbers} and {rationalnumbers}
C. {even numbers} and {odd numbers}
D. {multiples of five} and {multiples of ten}

Exercise 2: Sets (Venn Diagrams) What is the definition of the shared area in the following Venn Diagram.



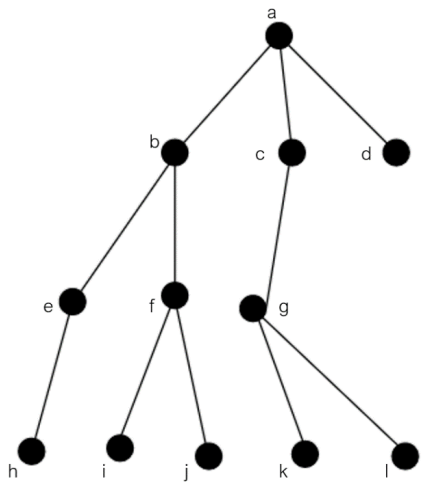
Exercise 3: Graphs

Draw the graph ({a,b},{(a,a),(b,b)}). How many unconnected subgraphs are there? Write them down in a mathematically correct form. Is this a directed, or undirected graph?

Exercise 4 Consider the following tree with 12 nodes.

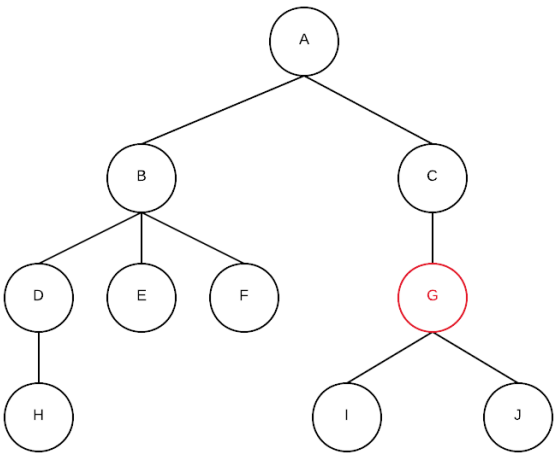
Answer the following questions:

a. Which of the nodes have two children?
b. Which of the nodes have no ancestors?
c. Which are the leaves on level 2 of the tree?
d. What are the children of the root?



Exercise 5:
Let the following search tree, with an initial state A and a state G that satistifes the goal conditions.

```
function TREE-SEARCH(problem,fringe)
return a solution or failure
        fringe
INSERT(MAKE-NODE(INITIAL-STATE[problem]), fringe)
        loop do
                if EMPTY?(fringe) then return failure
                node  REMOVE-FIRST(fringe)
                if GOAL-TEST[problem] applied to STATE[node] succeeds
                        then return SOLUTION(node)
                fringe
INSERT-ALL(EXPAND(node, problem), fringe)
```



Remember the algorithm for tree-search introduced in the lecture. Please follow the algorithm precisely (as if you were a Computer), given the following instructions. All the notes are added in inverse-alphabetic order in one group to the fringe (i.e. if node A is

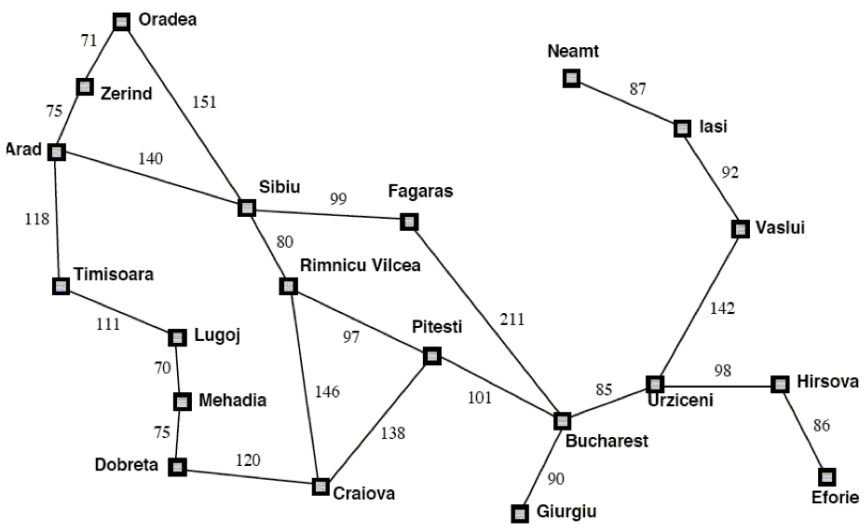
explored you add (C, B) to the fringe, either at the beginning if it is a FIFO queue, or at the end, if it is a LIFO list.

How many nodes do you need to explore before you find the goal state with DFS, BFD and IDFS? We count a node as explored when it is taken out of the fringe by the REMOVE-FIRST function.

Exercise 6: Informed search

Below you see a map of Romania. On the left connections are depicted with the actual distances. On the right you find information about the straight-line-distance from that town to Bucharest.

Hieronder vindt je een schematische kaart van Roemenië.

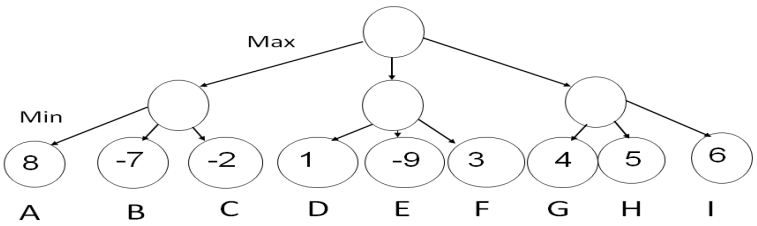


Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	176
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	10
Rimnieu Vikea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

a. Show how a **hill-climbing** algorithm would search a route from Arad to Bucharest, where the straight-line-distance is used to estimate the town nearest to Bucharest. Support your answer with a search tree.
b. Do exactly the same as in question a), but now for a A/A* algorithm.
c. Is the applied algorithm in question b) an A or A* algorithm? Explain why.

Exercise 7:

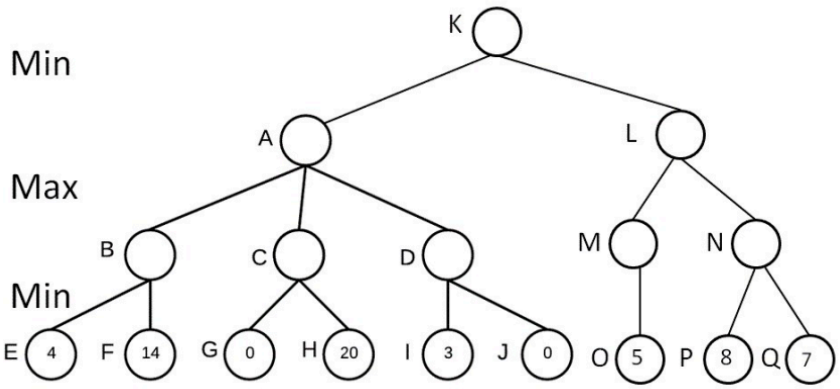
Given is the following search tree:



What is the expected value on the root of the tree if Max has the first move? Which nodes can you prune if you use Alpha/Beta pruning?

Exercise 8: Alpha/beta pruning 2

If you apply MiniMax with Alpha-beta pruning, which nodes in the tree can you ignore (prune). More than one answer can be correct.



a. H,J
b. N,P,Q
c. D,I,J
d. H,I
e. M,O
f. L,M,N,O,P,Q

Exercise 9: Truth tables
Complete the following truth table

A	B	C	A v -B	- (A v -B)	C-> - (A v -B)
True	True	True			
True	True	False			
True	False	True			
True	False	False			
False	True	True			
False	True	False			
False	False	True			
False	False	False			

Exercise 10: Entailment
Check whether the formula A -> (-A v B) entails A & B.

Exercise 11: Clause Normal Form (CNF)
Translate the formula (A -> (-A v B)) -> -B into Clause Normal Form.

Exercise 12: DPLL for SAT solving
Given a clause set
S = { A v B, -A v -B, A v -B v C, -A v B, A v -C}
Prove satisfiability, or unsatisfiability, with DPLL. Apply the MOMs heuristics, and assign the value 0 if the "winning literal" occurs with more negation signs than without, and 1 otherwise (equal or less). In case of a draw use alphabetical order, and assign 0 first.

Exercise 13: DPLL for Entailment

Prove, or disprove whether **C -> -(A v -B)** and **A** entail **-C** or not, using DPLL with alphabetical clause selection, and negation first.